Mathematical overview

The simultaneous gap map is D(a,b,c,d)=(|b-a|,|c-b|,|d-c|,|a-d|). Adding a common constant, or replacing each height h by M-h with a common ceiling M, leaves the next gap ring unchanged. These operations do not explain every collision: normalized nonnegative predecessors of (1,1,1,1) include both (0,1,0,1) and (0,1,2,1). For a target ring g, choose a sign for each gap and require the signed changes to sum to zero around the closed ring; cumulative sums, translated to minimum zero, give every normalized predecessor. Labels stay fixed. Heights 0-3 and at least one zero are constraints on printed old rings, not on later dynamics.

For binary heights, D compares each current old height with the next old height in numbered cyclic order, wrapping the last station to 1: equal gives 0, different gives 1. This is adjacent exclusive-or. Two rounds give the exclusive-or of entries two steps apart, four rounds entries four steps apart, and eight rounds entries eight steps apart. On an eight-station cycle the latter are the same entry, hence every output is zero. The six-station start (1,0,0,1,0,0) enters a period-three nonzero cycle. Thus this termination conclusion depends on station count. Children need no algebraic modulo notation to test and explain the doubling-distance rule.

With the directed rule E_i(v)=v_(i+1)-v_i, negative quantities are allowed. This is a different map from the absolute gap rule. The alternating start (0,1,0,1) becomes (1,-1,1,-1), then (-2,2,-2,2), and magnitudes double thereafter. It is unbounded. Every directed output sums to zero, but zero total does not bound magnitudes or imply termination. The base four-tower bound used absolute values and nonnegative heights; it cannot be carried over.

Grades 2-5 can search backward with cubes or cards; grades 4-5 can investigate six/eight binary dynamics and explain doubling; directed signed work suits children already comfortable with quantities below zero, roughly 4-5. Catalogs prove completeness only within their stated bounds. Experiments on a few eight-rings suggest the theorem; the doubling-distance argument proves it for all binary starts.

Entry route, materials, launch, and timing

Unpiloted. Mats, diagrams and trajectories are digitally checked; cube fit, old/new handling and signed-card procedures have not been physically tested. All work uses one unchanged reference state until a whole new state is built. Page 1 requires small absolute differences and understanding backward search. Page 2 requires binary matching, numbered cyclic positions, and retained old/new rows. Page 3 requires subtraction that may be negative. Offer the base cube-gap work to K-1; this companion does not force a young-child signed version.

Per pair: two copies of the packet or equivalent old/new mats, sixteen reversible 0/1 counters, eight erasable number cards, dry-erase marker, pencil, and scrap paper. Twelve stacking cubes optionally support the backward tower search. Large page-1 stations are 20.5 mm square; page-2 stations are 21.9 mm square. Small worked examples and recording tables are not manipulative mats. For page 2, cover the other station-count board while working, so the reference state is clear.

For eleven children in K–1 / 2–3 / 4–5, use five pair kits, eighty reversible counters, forty erasable cards, five markers, five pencils, and optionally sixty cubes. The third older child reads edges and checks outputs, rotating into the pair. Keep three adults anchored, one per table; young children can revisit base material. No printed set of all 256 eight-rings is required. If complete-case checking is chosen, pool cases and retain checkable cards; the general explanation is shorter.

Common launch, 4 minutes: handle a pair of small towers, then follow page 1's non-task input (1,3,2,0). A's new gap is 2; complete the whole output (2,1,2,1) before moving the old towers. For page 2, keep the same next-station convention around 6 or 8. For page 3, use its visual (1,3,3,1) -> (+2,0,-2,0); point to an up-gap and a down-gap. Move a marker two steps below zero on a number line if signed quantities are unfamiliar; do not make that conversion the whole investigation.

Suggested return visits: 20-30 minutes for predecessor classification, 20-40 for binary cycles and the eight-round explanation, and 15-30 for directed growth. A first route may stop with a complete predecessor catalog. Let deeper proof questions remain available beyond one hour. Keep one sequence of full states; rounds can be counted afterward.

Solutions, optional hints, extensions

1. For target (1,1,1,1), all six normalized old rings are (0,1,0,1),(0,1,2,1),(1,0,1,0),(1,0,1,2),(1,2,1,0),(2,1,0,1). For target (1,2,1,2), all four are (0,1,3,2),(1,0,2,3),(2,3,1,0),(3,2,0,1). Signed unit changes must include two rises and two falls for the first target: six sign choices. For the second, the signed gaps must cancel; there are four choices of a 1-gap and a 2-gap to rise, with the other two falling. Cumulative sums determine the relative heights and the minimum-zero condition fixes translation. Every resulting start fits 0-3. Rotation counts as different when the labeled tuple differs. Optional hint: walk around a prospective old ring and ask how much rise must be balanced by fall. Extension: remove the minimum-zero condition and classify all common translations within a fixed height ceiling; discuss complements and collisions outside those symmetries.

2. Six-station run: 100100 -> 101101 -> 110110 -> 011011 -> 101101; the last three states cycle forever. Eight-station single-1 start lasts exactly eight rounds: its seventh state is 11111111 and eighth is 00000000. After two rounds, each intermediate neighbor appears twice in the comparison and cancels, leaving a comparison two stations away. Repeat for four and eight rounds; eight stations away is the same station, so all outputs are zero. Optional hint: compare one output after two rounds and identify which old sites still matter. Extension: explore 5,7, or 16 stations; powers of two give the same doubling-distance cancellation, while a finite sample on another size does not prove a universal claim. Do not assert that every six-start cycles nontrivially; constant starts vanish immediately.

3. First run: (0,1,0,1) -> (1,-1,1,-1) -> (-2,2,-2,2) -> (4,-4,4,-4) -> (-8,8,-8,8) -> (16,-16,16,-16). It exceeds distance 10 from zero in round five. For an alternating row (a,-a,a,-a), every next gap is alternating with magnitude 2|a| and opposite signs. Repeated doubling beats any fixed positive ceiling. Second run through five rounds: (0,0,1,1), (0,1,0,-1), (1,-1,-1,1), (-2,0,2,0), (2,2,-2,-2), (0,-4,0,4). It does not exceed distance 10 by round five, so the printed question, "by round 5" (round 5 included, as the table's row numbers show), selects the first start. It offers a contrasting pattern; children need not prove a universal growth theorem for every nonconstant signed start. Optional hint: compare just the alternating pattern between consecutive rounds. Extension: investigate the unchanged total zero after a directed round and why it gives no magnitude bound.

New versus base, sources, check evidence

Base Week 49 asks whether selected predecessor rings exist, compares three/four stations, and studies parity, positive scaling and termination. Page 1 classifies every bounded normalized predecessor rather than merely supplying one; page 2 changes to six/eight binary cycles; page 3 changes to directed signed differences and loss of boundedness. These are three investigations in one theme.

Original extensions to the base source Daniel Shapiro, The Four-Number Game (2005), pp. 1-3. No new downloaded puzzle copied. Week 1 encore and Math Circle by the Bay preface, printed pp. viii-x (PDF pp. 9-11), support flexible depth. student/verify.py enumerates all 256 bounded four-starts, checks both worked examples and catalogs, checks all 256 binary eight-starts, and recomputes the six-cycle and both directed trajectories. General station-count and unboundedness explanations are separate from finite checks. No classroom or physical test is claimed.